

MARK ABRAHAM MAGUMBA

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1. Overview:

I am currently an active researcher in the field of sustainable and efficient AI, model compression, generative AI, natural language processing, and machine learning. I am also engaged in related areas like data science, Natural Language Processing, and machine learning in general. I am particularly interested in practical applications of data science and machine learning in finding novel solutions to various problem areas like health, nature conservation, sociology, agriculture, urban planning, and energy.

2. Current positions

- Postdoctoral Researcher – Research and Innovation, Technological University, Dublin, Ireland, September 2023 to date

3. Previous work experience

- Lecturer, Makerere University College of Computing and Information Sciences, Department of Information Systems, July 2022 to December 2025
- Assistant Lecturer, Makerere University College of Computing and Information Sciences, Department of Information Systems, August 2012 to July 2022
- ICT Consultant, National Forum for People Living with HIV AIDS Networks in Uganda, January 2013 to January 2014
- Computer Science Lecturer Kampala University, Uganda, June 2011 to June 2012
- ICT Manager Movit Products Limited, Uganda, June 2010 to April 2011

4. Education and degrees awarded

- PhD in Information Systems, Makerere University (Uganda), Areas: Knowledge Engineering, Machine Learning and Natural Language Processing in Health Informatics, titled ***Ontology Based Language Modelling for Generalized Automated Disease Incidence Detection in The Social Web***, (Graduated May 2021)
- MSc IT management, University of Sunderland (United Kingdom), Information Technology and Management, December 2009
- BSc Information Technology, Makerere University (Uganda), Information Technology, September 2007

5. Teaching Experience

I have taught several undergraduate and graduate courses including:

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| • Machine Learning (undergraduate) | • Applied Business Statistics (undergraduate) |
| • Artificial Intelligence (undergraduate) | • Web programming (undergraduate) |
| • Database Modelling and Development (undergraduate) | • Data Visualization (Graduate) |
| • Information Systems Security and Risk Management (undergraduate) | • Research Methodology (Graduate) |
| | • Network Security (Graduate) |

6. Research, Grants, prizes and honors

- Sep 2022 – Mar 2024: Principal investigator for the Google Research collab project entitled, **“BASIS: Broad Avian Species Surveillance with Intelligent Sensing”**
- Aug 2021 – Sep 2022: Co-Principal Investigator for the project entitled, ***Tracking COVID-19 Critical Care Resources***
- Aug 2021 – Sep 2022: Co-Principal Investigator for the project entitled, ***RAPID: A Real Time Application for Screening Symptoms of Perinatal Depression Among Expectant Mothers***
- Aug 2020 – Sep 2022: Co-Investigator for the project entitled ***eHealth Adoption in Uganda. What is the Status (eHAU)?***
- Mar 2020 – Aug 2020: Makerere RIF Special COVID-19 research and innovation award for the project entitled, **Data-Sanitizer: Semi-Automated Identification and Factchecking of Viral COVID-19 Related Posts Using Online Expert Panels**
- 2020 – Aug 2021: RISE 2019 COCIS seed grants award for the project entitled, ***An AI Powered Tool for Generalized Disease Surveillance using Unstructured Social Data***
- September 2018: Google travel grant to attend the 2018 International Conference on Data Science ICDATA'18 in Las Vegas, Nevada, USA
- July 2017: Best presentation award in Data Mining Session of the 2nd IEEE International Conference on Computational Intelligence and Applications (ICCIA 2017) in Beijing China • November 2014: NORHED/Hi-TRAIN PhD scholarship

7. Selected Publications

- **Magumba, Mark Abraham**, Hasifah Kasujja, and Tonny Justus Oyana. ***Neural Semantic Prediction of Prenatal EPDS subscale 3A Categories Using Socio-Demographic Features.*** IEEE Access (2025).
- **Magumba, Mark**, Steven Davy, and Owais bin Zuber. ***N for Parameter: Efficient Multi-Scale Neural Audio Super Resolution with GAN.*** ECAI 2025. IOS Press, 2025. 313-321.
- Gichuhi, Haron W., **Mark Magumba**, Manish Kumar, and Roy William Mayega. ***A machine learning approach to explore individual risk factors for tuberculosis treatment non-adherence in Mukono district.*** PLOS Global Public Health 3, no. 7 (2023): e0001466.
- **Magumba, Mark Abraham**, and Peter Nabende. ***Evaluation of different machine learning approaches and input text representations for multilingual classification of tweets for disease surveillance in the social web.*** Journal of Big Data 8, no. 1 (2021): 1-17.
- **Magumba, Mark Abraham**, Peter Nabende, and Ernest Mwebaze. ***Ontology boosted deep learning for disease name extraction from Twitter messages.*** Journal of Big Data 5, no. 1 (2018): 31.
- **Magumba, Mark Abraham**, Peter Nabende, and Ernest Mwebaze. ***Message Classification for Generalized Disease Incidence Detection with Topologically Derived Concept Embeddings.***, in the 2018 International Conference on Data Science, ICDATA'18, CSREA Press 2018
- **Magumba, Mark Abraham**, Peter Nabende, and Ernest Mwebaze. ***Ontology driven machine learning approach for disease name extraction from twitter messages.*** In 2017 2nd IEEE International Conference on Computational Intelligence and Applications (ICCIA), pp. 68-73. IEEE, 2017
- **Magumba, Mark Abraham**, and Peter Nabende. ***An Ontology for generalized disease incidence detection on twitter.*** In International Conference on Hybrid Artificial Intelligence Systems, pp:

38-51. Springer, Cham, 2017.

8. Professional Service

- Peer reviewer for SAGE Health Informatics Journal
- Panel member/ reviewer for HELINA (Health Informatics in Africa) 2021 Conference

9. Additional Training and Certifications

- AWS Certified Machine Learning – Specialty, AWS Training and Certification, January 2025
- AWS Certified Cloud Practitioner, AWS Training and Certification. August 2024
- Applied Data Science Module, Scientific Computing and Python for Data Science, WorldQuant University/ The Data incubator, September 30th, 2019 – 20th March 2020
- Open Science Grid (OSG) User School, University of Wisconsin-Madison (USA), July 2018
- Visiting PhD student at the University of Bergen (Norway), PhD taught courses (INF 283, INF 358, MNF 990), NORHED – HITRAIN project, August 2015 – December 2015
- OpenMRS Power User, Decimal Works, NORHED-HITRAIN project, February 2016
- Microsoft Sharepoint, Microsoft Innovations Centre/ Makerere University College of Computing and Information Sciences, September 2012
- Oracle Certified Associate 10g, Oracle Corporation, March 2010
- CIW (Certified international Webmaster), Pearson VUE, June 2009
- CIW professional, Pearson VUE, June 2009

9. Leadership and Other Activities

- Member of the Academic Steering committee, ISBAT University, Uganda (July 2012 to September 2012)
- Program Coordinator School of Computer Science and IT and Secretary to the Deans' Committee, Kampala University (August 2011 to July 2012)
- Minister of Finance of the Makerere University Computing and Information Technology Students' Society (2005/2006)
- Minister of Internal Affairs, Makerere University Livingstone Hall (2005/2006) • Chairman of the internal affairs committee, Makerere University Livingstone Hall (2005/2006)
- Makerere College School, Bikangaga House Prefect (2002/2003)
- President of St. Mary's College Kisubi Agriculture Club (2000/2001)